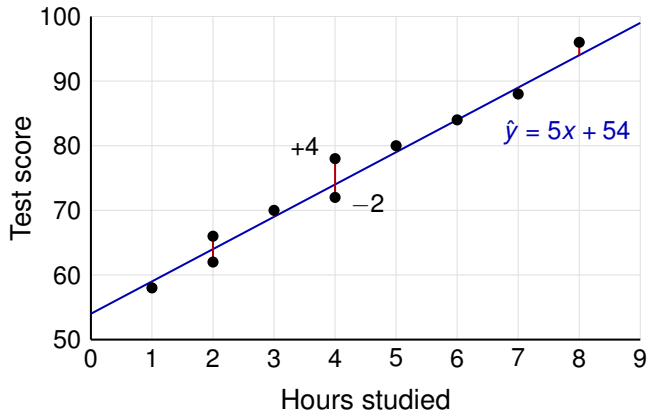
Residuals KEY

Name _____

A line of best fit makes predictions. Most real points are a little above or below the line. A **residual** measures how far off the prediction is.

residual = actual − predicted = $y - \hat{y}$
Positive residual: the point is **above** the line. Negative residual: the point is **below** the line.

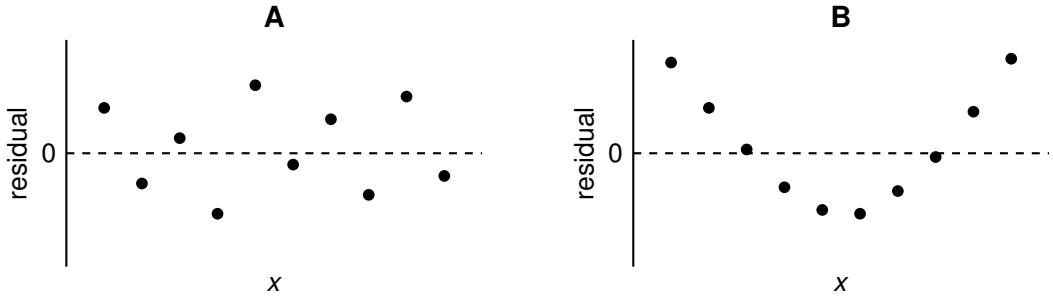


Example: The line is $\hat{y} = 5x + 54$. Two students studied 4 hours.

x	y (actual)	$\hat{y}$ (predicted)	$y - \hat{y}$	Meaning
4	78	$5(4) + 54 = 74$	$78 - 74 = 4$	scored 4 points **more** than predicted
4	72	74	$72 - 74 = -2$	scored 2 points **less** than predicted

The least-squares line. Square every residual and add them: $\Sigma(y - \hat{y})^2$. The **least-squares regression line** is the line that makes this sum as small as possible. Calculators and computers find this line for you.

Residual plots. Put x across the bottom and the residuals up the side.



A: random scatter, no pattern. A line is a **good** model.

B: a curved pattern. A line is **not** a good model. The data is curved.

Work with a partner. Compare answers when you finish each problem.

1. A plant's height (cm) is predicted by $\hat{y} = 2.5x + 4$, where x is the number of weeks. Fill in the table.

x (weeks)	y (actual)	$\hat{y}$ (predicted)	residual $y - \hat{y}$
1	6.8	6.5	0.3
2	8.6	9	-0.4
3	11.8	11.5	0.3
4	13.6	14	-0.4
5	16.9	16.5	0.4

   (a) Which weeks have points above the line? Weeks 1, 3, and 5 (positive residuals)

   (b) Is the point for week 2 above or below the line? How do you know? Below. Its residual is negative.

2. Use the study line $\hat{y} = 5x + 54$.

   (a) A student studied 6 hours and scored 90. Find the residual.
   $\hat{y} = 5(6) + 54 = 84$. Residual $= 90 - 84 = 6$.

   (b) Another student has a residual of -5. What does that mean?
   The student scored 5 points less than the line predicted.

3. The used car line is $\hat{y} = -2.5x + 30$ (value in thousands of dollars). A 6-year-old car is worth \$14,000. Find the residual and explain it.
   $\hat{y} = -2.5(6) + 30 = 15$. Residual $= 14 - 15 = -1$. The car is worth \$1,000 less than predicted.

4. Two lines were fit to the same 4 points. Here are their residuals.

	Residuals				$\Sigma(y-\hat{y})^2$
Line A	2	-1	-1	0	$4+1+1+0=6$
Line B	3	-2	1	-2	$9+4+1+4=18$

   (a) Find $\Sigma(y-\hat{y})^2$ for each line. (Square each residual, then add.)

   (b) Which line is closer to the least-squares line? Explain. Line A. Its sum of squared residuals is smaller.

5. Look at the residual plots on page 1.

   (a) Which plot shows that a line is a good model? A

   (b) What does the pattern in plot B tell you? The data is curved, so a straight line does not fit well.

6. A residual plot has positive residuals for small x, negative residuals in the middle, and positive residuals for large x. Is a line a good model? Explain.

   No. The residuals make a U-shaped pattern. The data is curved.